

HR Attrition Analytics Project

Excel | SQL | Python | Power BI

Project Overview

This project analyzes employee attrition patterns using Excel, SQL, Python, and Power BI. The objective is to identify key factors influencing employee turnover and generate business insights related to department performance, salary structure, employee satisfaction, age groups, and career stages.

Tools & Technologies Used

- Excel – Data cleaning, pivot analysis, dashboard preparation
- SQL – Business analysis queries and KPI extraction
- Python (Pandas, NumPy) – Data validation and analytical insights
- Power BI – Interactive HR attrition dashboard

Dataset Information

- Dataset Name: HR_Attrition_Data.csv
- Domain: Human Resources Analytics
- Project Type: Attrition & Workforce Analysis
- Key Columns: age, department, job_role, monthly_income, attrition, salary_band, age_group

Excel Analysis

- Performed data cleaning using Power Query.
- Validated missing values and duplicate records.
- Created pivot tables for department, salary, age, and job role analysis.
- Built KPI cards for attrition rate and employee count.
- Designed a clean HR Attrition Dashboard with slicers.

SQL Analysis

1. Data Import & Initial Validation
2. Core KPI & Attrition Analysis
3. Department & Job Role Analysis
4. Age, Experience & Career Analysis
5. Salary & Employee Satisfaction Analysis
6. Final Business Insights & Summary

Python Analysis

- Imported and cleaned HR dataset using Pandas.
- Validated missing values and duplicate records.
- Calculated attrition KPIs and workforce metrics.
- Analyzed department, salary, age, and satisfaction factors.
- Generated business insight logic using basic Python.

Power BI Dashboard

- Created interactive HR Attrition Dashboard.
- Added KPI cards for attrition rate.
- Designed department, role, salary, and age visuals.
- Implemented slicers for interactive filtering.
- Built executive-level business storytelling dashboard.

Key Business Insights

- Certain departments and job roles showed higher attrition rates.
- Lower salary bands experienced increased employee turnover.
- Younger employees displayed higher attrition tendency.
- Employee satisfaction and environment satisfaction influenced retention.
- Entry-level employees were more vulnerable to attrition.

Business Recommendations

- Strengthen employee retention strategies in high-risk departments.
- Improve salary structures for lower income groups.
- Introduce mentorship programs for early-career employees.
- Enhance workplace satisfaction and employee engagement.
- Monitor attrition trends regularly using dashboards.

Final Project Structure

```
HR_Attrition_Analytics_Project
|
├── data
|   └── HR_Attrition_Analysis.xlsx
├── Excel_Analysis
|   └── HR_Attrition_Analysis.xlsx
├── SQL_Analysis
|   └── hr_attrition_analysis.sql
└── Python_Analysis
```

```
|   └─ hr_attrition_analysis.py
└─ PowerBI_Dashboard
    └─ HR_Attrition_Dashboard.pbix
└─ output
```

Conclusion

The HR Attrition Analytics Project successfully identified workforce trends and attrition risk factors using multiple analytics tools. The project demonstrates practical business analysis skills, dashboard development, SQL analytics, and Python-based data analysis suitable for a professional analytics portfolio.